Mendelian Inheritance in *Drosophila melanogaster* Eye Color Crosses (F1 and F2 Analysis)

1 Introduction

Drosophila melanogaster, the common fruit fly, has been a cornerstone of genetic research for over a century. Its short generation time (10–14 days at 25°C), high fecundity, and easily distinguishable phenotypes make it an ideal model organism for studying inheritance patterns.

This experiment focuses on two classic eye color mutations:

- **White (w)**: An X-linked recessive mutation. The eyes are completely devoid of pigment, appearing white. The wild-type eye color is red.
- **Sepia (se)**: An autosomal recessive mutation. The eyes are a brownish-red color due to a defect in the conversion of a precursor molecule to bright red pigment.

By performing reciprocal crosses and analyzing both the F1 and F2 generations, you will learn to:

1. Distinguish between autosomal and sex-linked inheritance.
2. Apply the Chi-square (χ^2) statistical test to validate genetic hypotheses using F2 generation data.
3. Identify and discard contaminated or non-virgin crosses based on unexpected F1 outcomes.
4. Synthesize class data to make robust genetic inferences and account for experimental discrepancies.

2 Materials and Equipment

- **Fly Strains:**
  - Wild-type (red eyes, w^+w^+; se^+se^+)
  - White-eyed mutant (ww; se^+se^+)
  - Sepia-eyed mutant (w^+w^+; $sese$)

- **Equipment:**
  - Dissecting microscopes
  - Fly vials (25 x 95 mm)
  - Cotton plugs
  - Anesthetic: Ether (diethyl ether) and etherizers
  - Fly brushes or fine paintbrushes
  - White cards/sheets and permanent markers
  - Media: Standard cornmeal-molasses-agar *Drosophila* medium

3 Safety Precautions

1. **Ether is Highly Flammable**: Use in a well-ventilated area. Keep away from open flames, sparks, or hot surfaces.
2. **Ether is an Irritant**: Avoid skin contact and inhalation. Use the etherizer wand as instructed. If you feel dizzy, inform your TA immediately and move to fresh air.
3. **Hygiene**: Wash your hands thoroughly before and after handling flies or media to prevent contamination.

4 Experimental Design: Reciprocal Crosses

Your class will be divided into groups. Each group will be assigned one of the four reciprocal cross setups. Each group will set up three replicate vials for their assigned cross. This replication helps account for individual variation in fly viability or setup errors and allows for identification of contaminated vials.

Table 1: Parental Crosses and Their Purpose

Cross Number	Parental (P) Cross	Purpose
1	Red-eyed female x White-eyed male	To test X-linked recessive inheritance pattern
2	White-eyed female x Red-eyed male	To test X-linked recessive inheritance pattern (reciprocal)
3	Sepia-eyed female x Red-eyed male	To test autosomal recessive inheritance pattern
4	Red-eyed female x Sepia-eyed male	To test autosomal recessive inheritance pattern (reciprocal)

5 Laboratory Protocol

5.1 Setting Up Parental Crosses (Day 1)

1. **Preparation**: Label three vials for your assigned cross. Include: Cross Type (e.g., "Red female x White male"), your group name, and the date.

2. **Anesthetization**: Using the etherizer, gently anesthetize virgin female flies and male flies of the appropriate parental strains.

   - **Critical Step – Virgin Females**: To ensure that all offspring result from your cross, you must use virgin females. Collect females within 6–8 hours of eclosion (emergence from the pupa) before they have had a chance to mate. Their abdomens will appear smaller and pale compared to mated females.

3. **Identification**: Place the anesthetized flies on a white index card under a dissecting microscope. Use a fine brush to separate males from females and confirm their phenotypes.

   - **Sexing Flies**:
     - **Males**: Smaller abdomen with a dark, rounded tip. Presence of sex combs (small black bristles) on the front legs.
     - **Females**: Larger abdomen with a pointed, striped tip. No sex combs.

4. **Transfer**: Transfer the selected parental flies (2 males and 2 females) into each of your three replicate vials. Gently tap the vial to ensure the flies fall onto the media. Plug the vial with a cotton ball or foam stopper.

5. **Incubation**: Place vials in a 25°C incubator. Record the date of setup.

5.2 Removing Parents (Day 4–5)

After 4–5 days, the parents should have mated and laid eggs. You will begin to see larvae crawling on the media.

1. Anesthetize the flies in the vial.

2. **Critical Step**: Remove and discard all parental flies. They can still be fertile and will continue to lay eggs, making your F1 generation difficult to isolate.

3. Return the vials to the incubator.

5.3 F1 Generation Data Collection (Day 12–14)

The F1 generation will begin to eclose (emerge as adults).

1. Anesthetize the flies in the vial.

2. Using a brush, separate and count all flies. Record the number of:

- Red-eyed males
- Red-eyed females
- White-eyed males (if applicable)
- White-eyed females (if applicable)
- Sepia-eyed males (if applicable)
- Sepia-eyed females (if applicable)

3. **Expected F1 Phenotypes by Cross**:

Table 2: Expected F1 Phenotypes

Cross Number	Parental (P) Cross	Expected F1 Phenotypes
1	Red-eyed female x White-eyed male	All females: red; All males: red
2	White-eyed female x Red-eyed male	All females: red; All males: white
3	Sepia-eyed female x Red-eyed male	All red-eyed (males and females)
4	Red-eyed female x Sepia-eyed male	All red-eyed (males and females)

Note: Chi-square analysis is not performed on F1 data when only one phenotypic class is expected (Crosses 3 and 4) or when the pattern is strictly diagnostic (Crosses 1 and 2). The F1 generation serves primarily as a quality control step.

4. **Quality Control – Discarding Contaminated Vials**:

If any vial from your group shows F1 offspring that do not match the expected pattern above (e.g., a white-eyed female in Cross 1, a sepia-eyed fly in Cross 3, or any deviation from the expected sex-specific pattern), discard the data from that vial. Such deviations indicate that the parental female was likely not virgin, that a labeling error occurred, or that contamination from another stock took place. Do not carry such vials forward to the F2 generation. Record in your lab notebook which vials were discarded and why.

5. **Setting Up F1 Intercross (Day 12–14)**:

For vials that passed quality control, you will now set up the F1 intercross to generate the F2 generation.

(a) Anesthetize the F1 flies from a passing vial.

(b) Transfer 2 F1 females and 2 F1 males into a new, labeled vial with fresh media. The label should indicate the cross, generation (F1 intercross), and date. Set up one F2 vial from each F1 vial.

(c) Return the new vial to the incubator. The original F1 vial can be discarded after data recording.

5.4 F2 Generation Data Collection (Day 24–28)

The F2 generation will begin to eclose approximately 10–14 days after setting up the F1 intercross.

1. Anesthetize the flies in the F2 vial.
2. Using a brush, separate and count all flies. Record the number of all phenotypic classes present (red-eyed males, red-eyed females, white-eyed males, white-eyed females, sepia-eyed males, sepia-eyed females).
3. These F2 counts will form the basis of your statistical analysis.

6 Expected Patterns of Inheritance

6.1 Autosomal Recessive: Sepia Eye (se) × Red Eye (Wild-Type)

The sepia mutation is located on an autosome (chromosome 2). The wild-type red-eye allele (se^+) is dominant over the sepia allele (se).

Cross 3: Sepia-eyed female ($sese$) x Red-eyed male (se^+se^+)

Table 3: Cross 3 Inheritance Pattern

Generation	Genotypic Cross	Expected Phenotypic Ratio	Rationale
P	$sese \times se^+se^+$	Parental Phenotype: All sepia female; All red male	Parental
F1	All F1: se^+se	**F1 Phenotype:** All red-eyed (males and females)	All offspring are heterozygous, expressing the dominant red-eye allele
F1 Intercross	$se^+se \times se^+se$	**F2 Phenotype:** 3 red : 1 sepia (both sexes equally)	Monohybrid cross of heterozygotes yields the classic 3:1 Mendelian ratio. No sex linkage.

Cross 4: Red-eyed female (se^+se^+) x Sepia-eyed male ($sese$)

Table 4: Cross 4 Inheritance Pattern

Generation	Genotypic Cross	Expected Phenotypic Ratio	Rationale
P	$se^+se^+ \times sese$	Parental Phenotype: All red female; All sepia male	Parental
F1	All F1: se^+se	**F1 Phenotype:** All red-eyed (males and females)	Identical to Cross 3 F1. Autosomal inheritance produces identical results regardless of which parent carries the mutation.
F1 Intercross	$se^+se \times se^+se$	**F2 Phenotype:** 3 red : 1 sepia (both sexes equally)	Identical to Cross 3 F2. Reciprocal crosses yield the same F1 and F2 outcomes, confirming autosomal inheritance.

6.2 X-Linked Recessive: White Eye (w) × Red Eye (Wild-Type)

The white mutation is located on the X chromosome. The wild-type red-eye allele (w^+) is dominant over the white-eye allele (w). Males are hemizygous (carry only one X chromosome).

Cross 1: Red-eyed female (w^+w^+) **x White-eyed male** (w/Y)

Cross 2: White-eyed female (ww) **x Red-eyed male** (w^+/Y)

Key Diagnostic Difference: The reciprocal crosses in X-linked inheritance yield different F1 and F2 outcomes, unlike autosomal inheritance where reciprocal crosses yield identical outcomes.

7 Data Tabulation

7.1 Individual Group Data – Quality Control

Record your F1 counts for each replicate vial. Mark any vial that does not match expected F1 phenotypes as DISCARDED.

Example: Cross 1 (Red female x White male) – Your Group's Data. Note that this is only an example.

Table 5: Cross 1 Inheritance Pattern

Generation	Genotypic Cross	Expected Phenotypic Ratio	Rationale
P	$w^+w^+ \times w/Y$	Parental Phenotype: All red female; All white male	Parental
F1	Females: w^+w (red); Males: w^+/Y (red)	**F1 Phenotype:** All females: red; All males: red	Females inherit w^+ from mother and w from father (heterozygous red). Males inherit w^+ from mother (hemizygous red).
F1 Intercross	w^+w (female) $\times$ w^+/Y (male)	**F2 Phenotype:** All females: red; Males: 1 red : 1 white	Females: 100% red (w^+w or w^+w^+). Males: 50% red (w^+/Y), 50% white (w/Y). Overall: 2 red females : 1 red male : 1 white male.

Table 6: Cross 2 Inheritance Pattern

Generation	Genotypic Cross	Expected Phenotypic Ratio	Rationale
P	$ww \times w^+/Y$	Parental Phenotype: All white female; All red male	Parental
F1	Females: w^+w (red); Males: w/Y (white)	**F1 Phenotype:** All females: red; All males: white	Females inherit w^+ from father and w from mother (heterozygous red). Males inherit w from mother (hemizygous white).
F1 Intercross	w^+w (female) $\times$ w/Y (male)	**F2 Phenotype:** Females: 1 red : 1 white; Males: 1 red : 1 white	Females: 50% w^+w (red), 50% ww (white). Males: 50% w^+/Y (red), 50% w/Y (white). Overall: 1 red female : 1 white female : 1 red male : 1 white male.

Table 7: Example F1 Quality Control Data

Replicate	Red female	Red male	White female	White male	Expected?	Action
Vial 1	28	0	0	24	Yes	Carry to F2
Vial 2	0	31	0	29	No	**DISCARD**
Vial 3	26	0	0	27	Yes	Carry to F2

7.2 Class Data Compilation

Your instructor will compile the counts from all vials across the class for the four crosses. This is the dataset for your statistical analysis.

Table 8: F2 Data Table for Entire Class Data

Cross	Parental Cross	Total F2 Progeny (summing across entire class)
1	Red female x White male	Red female: ______, Red male: ______, White female: ______, White male: _____
2	White female x Red male	Red female: ______, Red male: ______, White female: ______, White male: _____
3	Sepia female x Red male	Red: ______, Sepia: _____
4	Red female x Sepia male	Red: ______, Sepia: _____

(Note: Enter the data summed over entire class.)

8 Statistical Analysis: Chi-Square (χ^2) Goodness-of-Fit Test on F2 Data

To determine if your observed F2 data fits the expected Mendelian ratios, you will use the Chi-square test.

8.1 Formulating Null Hypotheses (H_0) for Each Cross

- **Cross 1 (Red female x White male)**:

  - H_0: The white-eye mutation is X-linked recessive. The F2 generation will show a ratio of 2 red females : 1 red male : 1 white male (no white females). The deviation from this expected ratio is due to chance.

- **Cross 2 (White female x Red male)**:
  - H_0: The white-eye mutation is X-linked recessive. The F2 generation will show a ratio of 1 red female : 1 white female : 1 red male : 1 white male. The deviation from this expected ratio is due to chance.

- **Cross 3 (Sepia female x Red male)**:
  - H_0: The sepia-eye mutation is autosomal recessive. The F2 generation will show a ratio of 3 red : 1 sepia, with no sex difference. The deviation from this expected ratio is due to chance.

- **Cross 4 (Red female x Sepia male)**:
  - H_0: The sepia-eye mutation is autosomal recessive. The F2 generation will show a ratio of 3 red : 1 sepia, with no sex difference. The deviation from this expected ratio is due to chance.

8.2 Calculating Chi-Square (χ^2)

The formula is:

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

Where O = observed count, E = expected count.

8.2.1 Worked Example for Cross 3 Class Data

The following is a worked example to demonstrate how to perform the Chi-square calculation. Use this as a guide when completing your own calculations.

Observed (O): Red = 582, Sepia = 195. Total N = 777.

Expected (E) under 3:1 ratio: Red = $777 \times 3/4 = 582.75$, Sepia = $777 \times 1/4 = 194.25$.

Table 9: Cross 3 Chi-Square Calculation (Worked Example)

Phenotype	**Observed (O)**	**Expected (E)**	**(O-E)**	**(O-E)^2**	**(O-E)^2/E**
Red	582	582.75	-0.75	0.5625	0.00097
Sepia	195	194.25	0.75	0.5625	0.00290
Total	777	777			**0.00387**

Calculated Chi-square value = 0.00387

Degrees of Freedom (df): df = (number of phenotypic classes) − 1 = 2 − 1 = 1.

Student Data Tables: Use the following tables to perform Chi-square calculations using your class data.

Table 10: Cross 3 Chi-Square Calculation (Student Data)

Phenotype	**Observed (O)**	**Expected (E)**	**(O-E)**	**$(O\text{-}E)^2$**	**$(O\text{-}E)^2/E$**
Red					
Sepia					
Total					

Calculated Chi-square value = _____________
Degrees of Freedom (df): df = (number of phenotypic classes) − 1 = 2 − 1 = 1.

Table 11: Cross 4 Chi-Square Calculation (Student Data)

Phenotype	**Observed (O)**	**Expected (E)**	**(O-E)**	**$(O\text{-}E)^2$**	**$(O\text{-}E)^2/E$**
Red					
Sepia					
Total					

Calculated Chi-square value = _____________
Degrees of Freedom (df): df = (number of phenotypic classes) − 1 = 2 − 1 = 1.

Table 12: Cross 1 Chi-Square Calculation (Student Data)

Phenotype	**Observed (O)**	**Expected (E)**	**(O-E)**	**$(O\text{-}E)^2$**	**$(O\text{-}E)^2/E$**
Red female					
Red male					
White male					
Total					

Calculated Chi-square value = _____________
Degrees of Freedom (df): df = (number of phenotypic classes) − 1 = 3 − 1 = 2.

Table 13: Cross 2 Chi-Square Calculation (Student Data)

Phenotype	**Observed (O)**	**Expected (E)**	**(O-E)**	**(O-E)^2**	**(O-E)^2/E**
Red female					
Red male					
White female					
White male					
Total					

Calculated Chi-square value = _____________
Degrees of Freedom (df): df = (number of phenotypic classes) – 1 = 4 – 1 = 3.

8.3 Interpreting the Result

1. Determine the critical value from the Chi-square table for the appropriate df and a significance level (α) of 0.05:
   - df = 1: critical value = 3.841
   - df = 2: critical value = 5.991
   - df = 3: critical value = 7.815
2. Compare your calculated χ^2 to the critical value:
   - If χ^2 ¡ critical value: Fail to reject H_0. The deviation from the expected ratio is not statistically significant and is likely due to chance.
   - If $\chi^2 \geq$ critical value: Reject H_0. The deviation is statistically significant, and the observed data do not fit the expected Mendelian ratio.

9 If Inferences Differ from Known Inheritance

Sometimes, your data or the class data may not perfectly fit the expected Mendelian ratios. If your results lead to an inference that contradicts the known inheritance (e.g., you observe a significant deviation from the 3:1 ratio in the sepia cross or unexpected white-eyed females in Cross 1 F2), consider these possible explanations:

1. **Experimental Error:**
   - **Non-Virgin Females in Parental Setup**: This is the most common error. If a female in Cross 1 was not virgin, her F1 generation would contain unexpected phenotypes (e.g., white-eyed females). If such a vial was not discarded and was carried to F2, it would produce skewed ratios. The quality control step in F1 is designed to catch this.

- **F1 Intercross Setup Error**: If too few F1 flies were used, or if the sex ratio in the F1 intercross was imbalanced, the F2 generation might not represent the full range of expected genotypes.
- **Incorrect Sexing or Phenotyping**: Mistaking a male for a female, or misidentifying sepia versus white eyes under the microscope, will introduce counting errors.
- **Parental Escapes**: If a parental fly was not removed and continued to lay eggs alongside the F1 generation, it would add an extra cohort that does not represent the true F2.

2. **Biological Factors:**

   - **Differential Viability**: The mutant allele might confer a slight disadvantage. For example, white-eyed flies might have lower viability or emerge later than red-eyed siblings. If F2 scoring occurred on a single day, slower-developing classes may be underrepresented. Sepia-eyed flies in particular are known to have slightly reduced viability compared to wild-type.
   - **Sex-Linked Lethality**: Rarely, mutations on the X chromosome can cause reduced viability in hemizygous males, leading to a lower than expected male count in X-linked crosses.
   - **Non-Mendelian Inheritance**: Phenomena such as meiotic drive (where one allele is preferentially transmitted) can distort expected ratios, though this is rare for these well-characterized markers.
   - **Environmental Factors**: Temperature fluctuations, overcrowding in vials, or poor nutrition can affect the timing of eclosion and survival rates of different genotypes.

3. **Statistical Chance:**

   - Even if your hypothesis is correct, there is a 5% probability (if $\alpha = 0.05$) that a true null hypothesis will be rejected due to random sampling error (Type I error). This is why using class data (larger N) is more powerful than using just your group's data, as it reduces the impact of random sampling fluctuations.

10 Grand Synthesis

10.1 Conclusion of Inheritance Patterns Based on Class Data

Based on the F2 class data and Chi-square analysis performed for each cross, complete the following conclusions:

- **Cross 1 (Red female x White male)**: Therefore, conclude that based on our data, the inheritance pattern of this cross is ______________ (autosomal / sex-linked / cannot be sure as data were significantly different from expected).

- **Cross 2 (White female x Red male)**: Therefore, conclude that based on our data, the inheritance pattern of this cross is ______ (autosomal / sex-linked / cannot be sure as data were significantly different from expected).
- **Cross 3 (Sepia female x Red male)**: Therefore, conclude that based on our data, the inheritance pattern of this cross is ______ (autosomal / sex-linked / cannot be sure as data were significantly different from expected).
- **Cross 4 (Red female x Sepia male)**: Therefore, conclude that based on our data, the inheritance pattern of this cross is ______ (autosomal / sex-linked / cannot be sure as data were significantly different from expected).

10.2 Synthesis: Reconciling Red Eye Color as Both Autosomal and Sex-Linked

The red eye color in *Drosophila melanogaster* presents an interesting genetic paradox. In Crosses 1 and 2 (involving the white mutation), red eye color behaves as a sex-linked dominant trait—its inheritance pattern follows the X chromosome, appearing in specific sex-based ratios depending on the direction of the cross. In Crosses 3 and 4 (involving the sepia mutation), red eye color behaves as an autosomal dominant trait—its inheritance pattern is independent of sex and follows the classic 3:1 Mendelian ratio in the F2 generation.

This apparent contradiction is resolved by understanding that eye color is not determined by a single gene but by a biochemical pathway involving multiple genes. The red phenotype represents the wild-type state, which is produced only when all genes in the pigment biosynthesis pathway are functioning correctly. A mutation in any of these genes can disrupt pigment production and result in a different eye color.

- The *white* gene encodes a transporter protein responsible for depositing pigment precursors into pigment granules. When this X-linked gene is mutated, no pigment is deposited, resulting in white eyes regardless of the status of other eye color genes.
- The *sepia* gene encodes an enzyme involved in converting precursor molecules into the bright red pigment. When this autosomal gene is mutated, the pathway is blocked at a different step, resulting in the accumulation of a brownish pigment (sepia), while the white gene product functions normally.

What this tells us about inheritance and the relationship of genotypes to phenotypes:

1. **One phenotype can arise from mutations in multiple genes.** Red eye color is not the product of a single "red eye gene" but rather the default wild-type outcome when all genes in a pathway are functional. Therefore, red eyes can appear to be inherited in different patterns depending on which mutant allele is being crossed with the wild-type.
2. **The same phenotype (red eyes) can follow different modes of inheritance.** When red-eyed flies are crossed with white-eyed mutants, the red trait is inherited in an

X-linked pattern because the white gene resides on the X chromosome. When red-eyed flies are crossed with sepia-eyed mutants, the red trait is inherited in an autosomal pattern because the sepia gene resides on an autosome. Thus, the mode of inheritance is a property of the specific gene being studied, not of the phenotype itself.

3. **Genotype determines phenotype, but multiple genotypes can produce the same phenotype.** A red-eyed fly can have any of the following genotypes with respect to these two genes:

   - w^+w^+; se^+se^+ (completely wild-type)
   - w^+w^+; se^+se (heterozygous for sepia, still red)
   - w^+w; se^+se^+ (heterozygous for white in females, still red)
   - w^+w; se^+se (heterozygous for both, still red)
   - w^+/Y; se^+se^+ (wild-type male)

   Only when a fly is homozygous (or hemizygous in males) for a recessive mutation in a gene essential for pigment production does the eye color deviate from red.

4. **Genetic pathways integrate products from both autosomal and sex-linked genes.** The pigment biosynthesis pathway in *Drosophila* requires the coordinated function of genes located on different chromosomes. This illustrates that inheritance patterns cannot be understood in isolation—they reflect the underlying molecular biology of how gene products interact to produce observable traits.